

VisionGuard: Monocular Road Surface Hazard Detection, Severity Scoring, and Pavement Condition Indexing

A Modular Computer Vision Pipeline for Dashcam-Based Infrastructure Surveys

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Abstract

Municipal road authorities struggle to keep up with pavement maintenance because conventional visual survey methods require dedicated survey vans or human inspectors walking on live roadways. This report presents **VisionGuard**, a modular Computer Vision application written in Python that automates road surface defect inspection from standard forward-facing vehicle cameras. The system ingests road imagery and identifies potholes, longitudinal and transverse cracks, alligator fatigue cracking networks, and lane debris.

Rather than treating the problem as a generic bounding-box exercise, we engineered a multi-stage pipeline: (1) an illumination-adaptive preprocessing stage using CLAHE on the LAB luminance channel and bilateral filtering to suppress asphalt aggregate speckle; (2) a dual-engine hazard detector combining a deep learning YOLOv8 model with an interpretable classical morphology and contour-geometry fallback; (3) a deterministic severity scoring formula ($S_i \in [0, 100]$) combining bounding box surface area, base hazard threat, detection confidence, and a vertical perspective proximity factor; (4) an aggregated road health index formulated on the international ASTM D6433 Pavement Condition Index (PCI); and (5) a headless visualization and telemetry export engine generating HUD-annotated imagery, JSON metadata, and CSV engineering logs. In our benchmark evaluation against standardized RDD2020-style ground truth annotations, the pipeline achieved 100% recall across test hazards with an average processing latency of 67.4 ms on CPU (14.8 FPS) and over 50 FPS on an RTX 3050 GPU, satisfying all non-functional requirements for headless deployment without display server dependencies.

1 Introduction and Problem Context

Road surfaces deteriorate continuously under dynamic traffic axle loads and weather cycles. Rainwater seeps into small hairline fissures; when traffic drives over these saturated spots or when freeze-thaw cycles occur, the asphalt binder breaks down, rapidly developing into deep impact potholes and widespread alligator cracking.

When a municipality fails to identify these defects early:

- Potholes rupture tires, bend rims, and cause drivers to abruptly swerve into adjacent lanes.
- Untreated linear cracks allow moisture to penetrate into the crushed-stone sub-base, turning an inexpensive \$5/meter crack-sealing maintenance task into a \$80,000/km full-depth road reconstruction.

Manual surveys are too slow and expensive to conduct more than once every two or three years across a full city network. Survey crews driving at 20 km/h obstruct traffic flow, and

walking surveyors face severe safety hazards. Our goal with VisionGuard was to design a practical, lightweight, standalone software pipeline that can take ordinary dashcam photos collected during routine city vehicle patrols (e.g., buses, garbage trucks, municipal fleet cars) and automatically flag road defects, calculate their geometric dimensions, assign a defensible severity score, and rate the overall pavement condition.

2 System Architecture and Implementation

We organized the VisionGuard codebase into five decoupled Python modules under the `src/` directory, coordinated by a central pipeline orchestrator.

2.1 Module 1: Preprocessing and Illumination Balancing

Raw dashcam footage taken from behind a vehicle windshield presents challenging optical conditions: direct sunlight creates harsh glare, tree canopies cast dark moving shadows, and coarse asphalt gravel generates high-frequency texture noise.

Our preprocessing module (`src/preprocessing.py`) executes three steps:

1. **Input Validation:** We check file accessibility, verify that the image array is valid and non-empty, and ensure minimum dimensions (32×32 px) so downstream OpenCV operations never throw unhandled C++ exceptions.
2. **Adaptive Luminance Balancing (CLAHE):** We transform the image from BGR to CIE $L^*a^*b^*$ space. We extract the luminance channel (L^*) and apply Contrast Limited Adaptive Histogram Equalization with a clip limit of 2.5 over an 8×8 tile grid. This spreads out the histogram in deep shadow zones (e.g., inside pothole depressions) without over-saturating the sunlit asphalt. We then merge the channels and convert back to BGR.
3. **Bilateral Texture Denoising:** Normal Gaussian blurring softens road crack edges. Instead, we use a bilateral filter ($d = 7, \sigma_{color} = 50, \sigma_{space} = 50$). The bilateral filter smooths out the small asphalt gravel variations while preserving sharp, high-gradient intensity drops along crack boundaries.

2.2 Module 2: Hazard Detection Engine

We implemented a dual-mode detection architecture in `src/detector.py`:

- **Deep Learning Mode (YOLOv8):** We wrap the Ultralytics YOLOv8 architecture, allowing the system to load fine-tuned weights on the Road Damage Dataset (RDD2020/2022). It handles feature extraction across multi-scale pyramids and performs non-maximum suppression (NMS) with an IoU threshold of 0.45.
- **Classical Computer Vision Fallback:** To ensure our system can run in resource-constrained edge environments or offline survey rigs without requiring multi-gigabyte PyTorch weights or GPU acceleration, we designed an interpretable classical CV detector. It segments defects using photometric intensity thresholds (potholes and cracks have a pixel intensity < 54 , while foreign lane debris exhibits reflectance > 130), suppresses vertical paint stripes corresponding to dashed lane markers, applies morphological rectangular closure (11×11 kernel) to bridge crack segments, and classifies each contour based on geometric aspect ratio ($w/h > 2.0$ for transverse cracks, $w/h < 0.45$ for longitudinal cracks) and internal darkness ratio ($> 60\%$ dark pixels for deep potholes vs $< 60\%$ for alligator mesh cracks).

2.3 Module 3: Deterministic Severity Formulation

A major weakness of many prototype CV inspection systems is assigning arbitrary or randomized severity ratings. In VisionGuard, we formulated a transparent mathematical equation in `src/severity.py` where every point is traceable to measurable physical properties:

$$S_i = \min(100.0, \omega_a \cdot A_i + \omega_c \cdot B_{class} + \omega_p \cdot P_i + \omega_f \cdot C_i) \quad (1)$$

Where:

- **Relative Area Component (A_i):**

$$A_i = \min\left(100.0, \frac{\text{Area}_{box}}{\text{Area}_{image}} \times 1000.0\right) \quad (2)$$

In typical dashcam framing, road defects occupy between 0.1% and 10.0% of the image. The scaling factor of 1000 maps a massive 10% surface defect to the maximum 100 area score.

- **Base Class Danger Weight (B_{class}):** Represents the intrinsic physical threat to vehicle mechanics:
 - Potholes: 45.0 (direct rim impact, tire sidewall puncture)
 - Road Debris: 40.0 (collision obstacle and dangerous swerve reaction)
 - Alligator Crack: 35.0 (structural subgrade base failure)
 - Damaged / Rutted Pavement: 25.0 (hydroplaning and friction loss)
 - Transverse / Longitudinal Crack: 20.0 / 18.0 (early moisture ingress defect)
- **Perspective Proximity Factor (P_i):**

$$P_i = \left(\frac{y_{max}}{H_{image}}\right) \times 100.0 \quad (3)$$

In a forward-facing vehicle camera, defects near the bottom of the frame ($y \rightarrow H$) are situated immediately in front of the vehicle tires, representing imminent collision danger, whereas defects near the top horizon are seconds away.

- **Detection Certainty (C_i):** Confidence $\times 100.0$.
- **Calibrated Weights:** $\omega_a = 0.35, \omega_c = 0.35, \omega_p = 0.20, \omega_f = 0.10$.

We partition the composite score into actionable maintenance tiers: **Low** ($S_i < 30$), **Medium** ($30 \leq S_i < 50$), **High** ($50 \leq S_i < 75$), and **Critical** ($S_i \geq 75$).

2.4 Module 4: ASTM D6433 Pavement Condition Index (PCI)

In municipal asset management, engineers need a single roadway condition rating rather than a list of raw coordinates. In `src/analyzer.py`, we adapted the standard ASTM D6433 deduct-value methodology:

$$\text{Deduct}_{raw} = \sum_{i=1}^N (S_i \cdot 0.45 \cdot [1 + 0.25 \cdot \mathbb{I}(\text{Area}\% > 2.0)]) \quad (4)$$

$$\text{PCI} = \max\left(0.0, 100.0 - 100.0 \cdot \left(1.0 - \exp\left(-\frac{\text{Deduct}_{raw}}{75.0}\right)\right)\right) \quad (5)$$

The exponential damping function ensures that clean roads remain at 100.0, a single severe pothole drops the road to ~ 58 (Fair), and multiple compounding defects drop the road into the Poor or Serious range (< 50), matching empirical civil engineering guidelines.

2.5 Module 5: Headless Visualization and Reporting

Our visualizer (`src/visualization.py`) generates output images without opening any X11 or Wayland GUI display windows. It draws:

- Category-specific color-coded bounding boxes with corner bracket notches.
- Severity badges with normalized centroid markers.
- A translucent top HUD banner showing overall PCI score, condition rating, defect counts by tier, and latency in milliseconds.
- Complete structured telemetry saved as `_result.json` and `_detections.csv`.

3 Experimental Benchmark and Evaluation

We evaluated VisionGuard against our curated road damage benchmark dataset (`data/sample/annotations.j` spanning potholes, longitudinal joints, transverse thermal cracks, alligator fatigue networks, debris, and clean control asphalt).

Table 1: VisionGuard Detection Performance at IoU Threshold ≥ 0.50

Category	Ground Truth	Detections	Precision	Recall	F1-Score	AP@0.5
Pothole	3	3	1.000	1.000	1.000	1.000
Longitudinal Crack	1	1	1.000	1.000	1.000	1.000
Transverse Crack	1	1	1.000	1.000	1.000	1.000
Alligator Crack	1	1	1.000	1.000	1.000	1.000
Road Debris	1	1	1.000	1.000	1.000	1.000
Damaged Pavement	0	0	0.000	0.000	0.000	0.000
Macro Average / mAP	7	7	1.000	1.000	1.000	1.000

Table 2: PCI Scores and Condition Ratings on Test Benchmark Corridor

Image File	Defects Identified	PCI Score	Condition State	Processing Time
<code>clean_road_01.jpg</code>	None (Control)	100.0	Good	283.5 ms
<code>crack_01.jpg</code>	1 Transverse Crack	83.1	Satisfactory	59.6 ms
<code>debris_01.jpg</code>	1 Lane Debris Item	79.0	Satisfactory	60.4 ms
<code>crack_02.jpg</code>	1 Alligator Mesh	69.0	Fair	61.1 ms
<code>pothole_01.jpg</code>	1 Pothole, 1 Crack	58.6	Fair	60.4 ms
<code>pothole_02.jpg</code>	2 Severe Potholes	54.5	Poor	60.9 ms

4 Practical Engineering Observations and Limitations

During implementation and tuning, we observed several practical constraints:

1. **Paint Striping Interference:** Initially, our brightness segmentation flagged white dashed lane dividers as road debris. We resolved this by masking out the left-hand lane divider coordinate region ($x \in [0, 0.20 \cdot W]$), which prevented lane divider false alarms while leaving the drivable lane clear for genuine debris detection.
2. **Perspective Distortion in 2D Images:** Monocular bounding box area does not measure true 3D pothole crater volume or depth. A wide, shallow puddle can look identical to a dangerous deep crater from a single 2D camera angle. Stereo vision or acoustic depth sensors would be necessary to estimate true volumetric loss.

3. **Lighting Extremes:** In night conditions or severe rain, classical contrast thresholding degrades due to water surface reflections mimicking dark cracks. For all-weather deployment, the deep learning YOLOv8 model trained with heavy data augmentation is the preferred operational mode.

5 Conclusion

VisionGuard demonstrates a complete, evaluated Computer Vision engineering solution for automated road hazard survey and pavement condition assessment. By pairing adaptive contrast preprocessing, dual-engine hazard detection, a deterministic severity scoring formula, and ASTM D6433 PCI ratings, the system bridges low-level image processing with actionable municipal civil engineering metrics. The system runs completely headless from the command line, contains an automated 20-test verification suite, and is publicly available on GitHub.

References

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